NHANES | HbA1c ML benchmarks

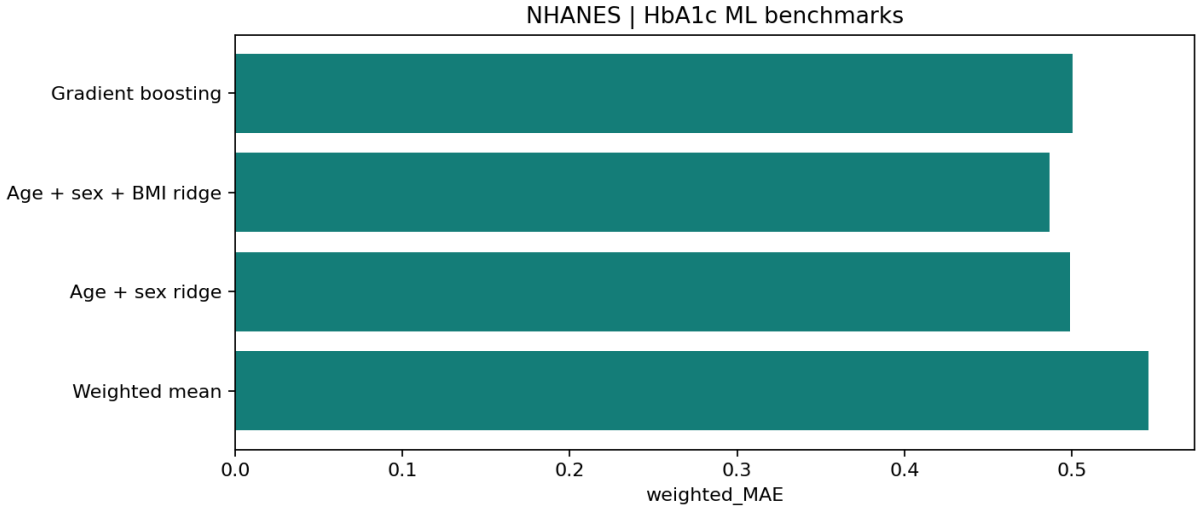
Scope

4,940 complete-case adults, NHANES 2017-2018. Predict concurrent HbA1c (%) using age, sex and BMI. Original survey association analysis remains the primary analysis.

Validation

Five outer folds hold out entire nested stratum/PSU groups. All transformations and weighted fits use training folds only. Hyperparameters were fixed in advance; no outer-fold model selection. Every adult receives one held-out prediction. Weighted metrics use MEC weights; these are predictive scores, not survey-design confidence intervals.

model	weighted_MAE
Weighted mean	0.5460
Age + sex ridge	0.4987
Age + sex + BMI ridge	0.4867
Gradient boosting	0.5003



Interpretation and limits

Cross-sectional prediction does not establish causality or predict future diabetes. Complete cases are a selected population; no external validation. Fold differences and sex/age errors are diagnostics, not proof of fairness. Race, diagnosis and outcome-derived features are excluded.